
GENERAL RELATIVITY

Tutorial problem set 11, 27.11.2020, with solutions

■ PROBLEM 29 Birkhoff's theorem.

The most general spherically symmetric line element is of the form

$$ds^2 = -A(t, r)dt^2 + 2B(t, r)dtdr + C(t, r)dr^2 + D(t, r)r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2) . \quad (29.1)$$

In problems (a) and (b) you will show that this line element can be reduced to a simpler looking one via coordinate transformation. Therefore, the simpler form of the line element will be just as general as (29.1).

(a) Show that the coordinate transformation of the form

$$r \rightarrow \bar{r} = \bar{r}(r, t) , \quad (29.2)$$

can be chosen such that (after suitable redefinitions) line element reduces to

$$ds^2 = -a(t, r)dt^2 + 2b(t, r)dtdr + c(t, r)dr^2 + r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2) . \quad (29.3)$$

(b) Show that the coordinate transformation of the form

$$t \rightarrow \bar{t} = \bar{t}(t, r) , \quad (29.4)$$

which eliminates the mixed term $dtdr$, always exists and thus the most general spherically symmetric line element (after suitable redefinition of functions) can be written in the form

$$ds^2 = -e^{\nu(t, r)}dt^2 + e^{\mu(t, r)}dr^2 + r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2) . \quad (29.5)$$

Hint: The non-vanishing components of the Riemann tensor for the line element (29.5) are given to you to be,

$$R^0_{101} = \frac{1}{4}e^{\mu-\nu} [2\partial_t^2\mu + (\partial_t\mu)^2 - \partial_t\nu\partial_t\mu] + \frac{1}{4} [\partial_r\nu\partial_r\mu - 2\partial_r^2\nu - (\partial_r\nu)^2] , \quad (29.6)$$

$$R^0_{202} = -\frac{1}{2}re^{-\mu}\partial_r\nu , \quad R^0_{303} = -\frac{1}{2}re^{-\mu}\sin^2\vartheta\partial_r\nu , \quad (29.7)$$

$$R^0_{212} = -\frac{1}{2}re^{-\nu}\partial_t\mu , \quad R^0_{313} = -\frac{1}{2}re^{-\nu}\sin^2\vartheta\partial_t\mu , \quad (29.8)$$

$$R^1_{212} = \frac{1}{2}re^{-\mu}\partial_r\mu , \quad R^1_{313} = \frac{1}{2}re^{-\mu}\sin^2\vartheta\partial_r\mu , \quad R^2_{323} = (1 - e^{-\mu})\sin^2\vartheta , \quad (29.9)$$

and the non-vanishing components of the Ricci tensor are R_{00} , R_{11} , R_{22} , R_{33} , and $R_{01} = R_{10}$. You do not need to calculate the Ricci scalar, why?

After casting the line element in a simpler form (but equally general), in the remaining problems you will determine what functions ν and μ are from the Einstein equation.

(c) Show that the tr component of the Einstein equation implies that μ is independent of time, $\mu(t, r) = \mu(r)$.

(d) Use (a linear combination of) the remaining components of the Einstein equation to show that

$$\nu(t, r) = -\mu(r) + f(t) . \quad (29.10)$$

(e) Use these results to conclude that the most general spherically symmetric solution to the Einstein equation in the vacuum (after a redefinition of the time coordinate) can be written as

$$ds^2 = -e^{-\mu(r)} dt^2 + e^{\mu(r)} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2) . \quad (29.11)$$

(f) Show that the other components of the Einstein equation imply

$$e^{-\mu(r)} = 1 - \frac{R_S}{r} , \quad (29.12)$$

where R_S is a real constant, so that we obtain the Schwarzschild solution.

Schwarzschild geometry is the most general asymptotically flat spherically symmetric solution to the Einstein equation. This is the Birkhoff-Jebsen theorem.

SOLUTION: This problem is (mostly) solved in section 5.2 of Carroll.

■ PROBLEM 30 Newton's potential.

Let us say you have some static spherical distribution of matter (a planet for example). Naturally, one would expect the solution for the metric of such a system to be time-independent, and to possess spherical symmetry (hence use spherical coordinates). Also, far away from the planet, one would expect the space-time to be almost Minkowski.

(a) Assume the following form for the metric,

$$ds^2 = -[1 + f(r)]dt^2 + [1 + w(r)]dr^2 + r^2 d\vartheta^2 + r^2 \sin^2(\vartheta) d\varphi^2 , \quad (30.1)$$

and derive the equations for f and w from the Einstein's equation.

(b) Show that the leading solution far away from the source reproduces the Newton's potential. Can you find the first correction to Newton's potential?

SOLUTION: (a) The nonvanishing Christoffel symbols are,

$$\Gamma_{rt}^t = \Gamma_{tr}^t = \frac{1}{2} \frac{f'}{1+f} , \quad (30.2)$$

et cetera, where $f' = df(r)/dr$ and $w' = dw/dr$.

If one knows the Christoffel symbols one can – in the usual way – calculate the nonvanishing components of the Riemann tensor. An easier way to obtain the Riemann tensor is from Eqs. (29.9), with the identifications $\nu(t, r) \rightarrow \ln(1 + f(r))$ and $\mu(t, r) \rightarrow \ln(1 + w(r))$,

$$R^0_{101} = \frac{1}{4} \left[\frac{f'w'}{(1+f)(1+w)} + \frac{f'^2}{(1+f)^2} - \frac{f''}{1+f} \right], \quad (30.3)$$

$$R^0_{202} = -\frac{r}{2} \frac{f'}{(1+f)(1+w)}, \quad R^0_{303} = -\frac{r}{2} \frac{f'}{(1+f)(1+w)} \sin^2(\vartheta), \quad (30.4)$$

$$R^1_{212} = \frac{r}{2} \frac{w'}{(1+w)^2}, \quad R^1_{313} = \frac{r}{2} \frac{w'}{(1+w)^2} \sin^2(\vartheta), \quad R^2_{323} = \frac{w}{1+w} \sin^2(\vartheta). \quad (30.5)$$

The nonvanishing Ricci tensor elements are then,

$$R_{00} = -\frac{f'^2}{4(1+f)(1+w)} - \frac{f'w'}{4(1+w)^2} + \frac{f''}{2(1+w)} + \frac{f'}{r(1+w)}, \quad (30.6)$$

$$R_{11} = \frac{f'^2}{4(1+f)^2} + \frac{f'w'}{4(1+f)(1+w)} - \frac{f''}{2(1+f)} + \frac{w'}{r(1+w)}, \quad (30.7)$$

$$R_{22} = -\frac{r}{2} \frac{f'}{(1+f)(1+w)} + \frac{r}{2} \frac{w'}{(1+w)^2} + \frac{w}{(1+w)}, \quad (30.8)$$

$$R_{33} = R_{22} \sin^2(\vartheta), \quad (30.9)$$

from which one obtains the Ricci scalar,

$$R = \frac{f'^2}{2(1+f)^2(1+w)} + \frac{f'w'}{2(1+f)(1+w)^2} - \frac{f''}{(1+f)(1+w)} - \frac{2f'}{r(1+f)(1+w)} + \frac{2w'}{r(1+w)^2} + \frac{2w}{r^2(1+w)}, \quad (30.10)$$

and the Einstein tensor,

$$G_{00} = \frac{1}{r} \frac{(1+f)w'}{(1+w)^2} + \frac{1}{r^2} \frac{(1+f)w}{(1+w)}, \quad (30.11)$$

$$G_{11} = \frac{f'}{r(1+f)} - \frac{w}{r^2}, \quad (30.12)$$

$$G_{22} = \frac{r^2}{4} \left[-\frac{f'^2}{(1+f)^2(1+w)} - \frac{f'w'}{(1+f)(1+w)^2} + \frac{2f''}{(1+f)(1+w)} \right] + \frac{r}{2} \left[\frac{f'}{(1+f)(1+w)} - \frac{w'}{(1+w)^2} \right], \quad (30.13)$$

$$G_{33} = G_{22} \sin^2(\vartheta) \quad (30.14)$$

Due to the Birkhoff theorem, one can approximate the energy momentum tensor of the star whose mass is M (at sufficiently large distances) can be written as,

$$T_{\mu\nu} = \delta_\mu^0 \delta_\nu^0 M c^2 \delta^3(\vec{x}), \quad (30.15)$$

such that the Einstein's equations are,

$$\frac{1}{r} \frac{(1+f)w'}{(1+w)^2} + \frac{1}{r^2} \frac{(1+f)w}{(1+w)} = \frac{8\pi GM}{c^2} \delta^3(\vec{x}) , \quad (30.16)$$

$$\frac{f'}{r(1+f)} - \frac{w}{r^2} = 0 , \quad (30.17)$$

$$\begin{aligned} \frac{r^2}{4} \left[-\frac{f'^2}{(1+f)^2(1+w)} - \frac{f'w'}{(1+f)(1+w)^2} + \frac{2f''}{(1+f)(1+w)} \right] \\ + \frac{r}{2} \left[\frac{f'}{(1+f)(1+w)} - \frac{w'}{(1+w)^2} \right] = 0 , \end{aligned} \quad (30.18)$$

$$G_{33} = G_{22} \sin^2(\vartheta) = 0 \quad (30.19)$$

(b) Linearizing equations (30.19) in f and w one obtains,

$$\frac{w'}{r} + \frac{w}{r^2} = \frac{8\pi GM}{c^2} \delta^3(\vec{x}) , \quad (30.20)$$

$$\frac{f'}{r} - \frac{w}{r^2} = 0 , \quad (30.21)$$

$$f'' + \frac{f' - w'}{r} = 0 , \quad (30.22)$$

where we dropped the (33) equation, as it is not an independent equation (it is linearly dependent on the (22) equation). Summing all three equations (30.20–30.22) gives,

$$f'' + \frac{2}{r} f' = \nabla^2 f(r) = \frac{8\pi GM}{c^2} \delta^3(\vec{x}) , \quad (30.23)$$

where ∇^2 is the Laplace operator. The solution of (30.23) is well known,

$$f(r) = -\frac{2GM}{c^2 r} \equiv \frac{2}{c^2} \phi_N(r) , \quad (30.24)$$

where $\phi_N(r) = -GM/r$ is the Newton's potential of a point mass M . Next, from Eq. (30.21) we see that,

$$w(r) = \frac{2GM}{c^2 r} = -f(r) \equiv -\frac{2}{c^2} \phi_N(r) . \quad (30.25)$$

For consistency, one can check that the solutions (30.24) and (30.25) satisfy the remaining Einstein's equation (30.22).

To get higher order (strong field) corrections to the potential, we have to expand the functions f and w . A convenient parametrization is,

$$f(r) = \lambda f_1 + \lambda^2 f_2 + \dots , \quad w(r) = \lambda w_1 + \lambda^2 w_2(r) + \mathcal{O}(\lambda^3) , \quad T_{\mu\nu} \rightarrow \lambda T_{\mu\nu} , \quad (30.26)$$

where λ is a perturbation parameter which can be set to *one* at the end of the calculation. Next, we can expand equations (30.19) in powers of f and w as,

$$\frac{1}{r} (1+f-2w)w' + \frac{1}{r^2} (1+f-w)w \simeq \lambda \frac{8\pi GM}{c^2} \delta^3(\vec{x}) , \quad (30.27)$$

$$\frac{(1-f)f'}{r} - \frac{w}{r^2} \simeq 0 , \quad (30.28)$$

$$\frac{r^2}{4} \left[-f'^2 - f'w' + 2(1-f-w)f'' \right] + \frac{r}{2} \left[(1-f-w)f' - (1-2w)w' \right] \simeq 0 . \quad (30.29)$$

Upon inserting (30.26) into (30.29) at the linear order in λ^1 one obtains equations (30.22), which we have already solved. At the quadratic order in λ the equations are then,

$$\frac{w'_2}{r} + \frac{w_2}{r^2} + \frac{1}{r}(f_1 - 2w_1)w'_1 + \frac{1}{r^2}(f_1 - w_1)w_1 \simeq 0, \quad (30.30)$$

$$\frac{f'_2}{r} - \frac{w_2}{r^2} - \frac{f_1 f'_1}{r} \simeq 0, \quad (30.31)$$

$$\frac{r^2}{4} [2f''_2] + \frac{r}{2} [f'_2 - w'_2] + \frac{r^2}{4} [-f_1'^2 - f'_1 w'_1 - 2(f_1 + w_1)f''_1] + \frac{r}{2} [-(f_1 + w_1)f'_1 + 2w_1 w'_1] \simeq 0, \quad (30.32)$$

or equivalently,

$$\frac{w'_2}{r} + \frac{w_2}{r^2} \simeq -\frac{f_1^2}{r^2} = -\frac{(2GM)^2}{c^4} \frac{1}{r^4} = -\frac{r_S^2}{r^4} \quad (30.33)$$

$$\frac{f'_2}{r} - \frac{w_2}{r^2} \simeq -\frac{f_1^2}{r^2} = -\frac{r_S^2}{r^4} \quad (30.34)$$

$$f''_2 + \frac{f'_2 - w'_2}{r} \simeq \frac{2f_1^2}{r^2} = \frac{2r_S^2}{r^4}, \quad (30.35)$$

where we made use of $f_1 = -w_1 = -2GM/(c^2 r)$, $f'_1 = -w'_1 = -f_1/r$, $f''_1 = 2f_1/r^2$, and $r_S = 2GM/c^2$.

Now, upon multiplying Eq. (30.33) by r^2 , it reduces to,

$$rw'_2 + w_2 = (rw_2)' = -\frac{r_S^2}{r^2}, \quad (30.36)$$

which can be easily integrated to give, $w_2 = r_S^2/r^2 + C_2/r$. The constant C_2 can be set to zero, since it naturally belongs to w_1 . In fact, the constant in w_1 can be uniquely fixed by the Gauss' integral, which was used to unambiguously determine the coefficient of the $1/r$ term in terms of the source strength GM/c^2 . With this in mind we have,

$$w_2 = \frac{r_S^2}{r^2}, \quad (30.37)$$

Inserting this into (30.34) gives,

$$\frac{f'_2}{r} = 0, \quad (30.38)$$

such that $f_2 = 0$. As above, the integration constant is uniquely fixed by the Gauss integral, which was used to determine f_1 . What remains to be done is to check whether (30.35) is satisfied. A trivial calculation shows that it is. As you have learned in the lecture, these solutions are the first two coefficients in the Taylor series of the Schwarzschild solution, for which $f = f_1 = -\frac{r_S}{r}$ and $w = \frac{r_S}{r} / (1 - \frac{r_S}{r})$.

■ PROBLEM 31 Mitchell's observation.

John Mitchell conjectured in 1783 the existence of objects with black-hole properties. (A few years later, Pierre-Simon Laplace suggested essentially the same idea.) Making use of Newtonian physics, he realized that for every massive object there exists a radius r_S , such that the escape velocity is greater than the speed of light for radial coordinates $r < r_S$. Find the value of this radius as a function of the mass M of the object, the gravitational constant G_N , and the speed of light c .

Do you notice any relation with the radial coordinates of the metric singularities in the Schwarzschild solution?

SOLUTION: We are interested in the escape velocity of a test particle of mass m . To calculate it, recall that in Newton's theory there is the conserved energy in the gravitational field of an object with a mass M ,

$$E = \frac{mv^2}{2} - \frac{GmM}{r}. \quad (31.1)$$

The escape velocity is defined such that at $r \rightarrow \infty$ particle's velocity vanishes, $v \rightarrow 0$, *i.e.* its conserved energy E in (31.1) is zero. From here we immediately get,

$$v_{\text{esc}}^2 = \frac{2GM}{r_S} \rightarrow c^2, \quad (31.2)$$

from where we obtain,

$$r_S = \frac{2GM}{c^2}, \quad (31.3)$$

which is equal to the Schwarzschild radius, at which the Schwarzschild solution is singular. The fact that a Newtonian calculation gives exactly equal answer to a general relativistic calculation is a coincidence (rigorously speaking, it is not exactly the same since the interpretation of escape velocity and the Schwarzschild radius is not the same). One would expect that a calculation such as the one above gives a radius which – up to a constant of the order unity – reproduces the Schwarzschild radius. The fact that this constant is equal to unity is a coincidence.